

Task - 5: Malware Types & Behavior Analysis (Basic)

Internship Task – Elevate Labs

This task focused on understanding different types of malware, their behavior, and basic analysis techniques. The goal was to gain awareness of common malware threats and how they can be detected and prevented.

1. Introduction to Malware

Malware refers to malicious software designed to damage systems, steal sensitive data, or disrupt operations. It targets computers, mobile devices, and networks.

2. Types of Malware

Virus: Attaches to legitimate files and spreads when executed.

Worm: Self-replicates and spreads automatically through networks.

Trojan: Disguises itself as a legitimate program to trick users.

Ransomware: Encrypts files and demands payment for recovery.

3. Malware Analysis Using VirusTotal

VirusTotal was used to analyze known malware samples by submitting file hashes. It provides detection results from multiple antivirus engines, helping identify malicious files.

4. Detection Reports and Behavior Indicators

Detection reports include: - Malware classification - Detection ratio - Threat indicators

Behavior indicators reveal how malware interacts with files, processes, and network connections.

5. Malware Lifecycle

Malware follows a lifecycle: - Infection - Execution - Persistence - Propagation - Damage or payload execution

6. Malware Spreading Techniques

Malware commonly spreads through: - Phishing emails - Malicious downloads - Infected websites - USB devices

7. Malware Prevention Methods

- Install and update antivirus software - Avoid suspicious links and downloads - Keep systems patched - Enable firewalls - Educate users on safe browsing

8. Summary

Malware analysis tools help detect threats early. Strong security practices and user awareness reduce malware risks significantly.

Learning Outcome

This task improved my understanding of malware types, behavior analysis, and prevention methods. I learned how malware spreads, how detection tools work, and how to protect systems from basic malware threats.